Unit

26 Homologous series, structural formulae and naming of carbon compounds

Unit Key Concepts

- Representation of structures of carbon compounds
- Homologous series and functional groups
- Systematic naming of carbon compounds

Pre-study

A pre-study worksheet on this unit is available for download from *Jing Kung Chemistry Website*.

26.1 A look at molecules of compounds in petroleum

The gaseous fraction from fractional distillation of petroleum contains compounds with low boiling points (less than 40 °C). The hydrocarbon molecules in this fraction contain 1 to 4 carbon atoms. On the other hand, the liquid fractions consist of molecules having from 5 to more than 25 carbon atoms.

Now complete the following discussion to learn more about the physical properties of hydrocarbons.

Worksheet

A worksheet on this activity is available for download from *Jing Kung Chemistry Website*.

Discussion

Table 26.1 shows the boiling points of some hydrocarbons.

Table 26.1

Boiling points of some hydrocarbons

Hydrocarbon	Molecular formula	Boiling point (°C)
Butane	C_4H_{10}	–0.5
Ethane	C_2H_6	–89
Heptane	C_7H_{16}	98
Hexane	C_6H_{14}	69
Methane	CH_4	–162
Octane	C_8H_{18}	126
Pentane	C_5H_{12}	36
Propane	C_3H_8	–42

Chemists often gather data about the physical properties of substances. These data can be organized in many ways. The most useful way is the one that tells us the trends among the data.

1 Propose a useful way to arrange the data in Table 26.1.

2 Reorganize the data table based on your proposal.

Discussion

1 Arrange the data in increasing number of carbon atoms in the molecular formula of hydrocarbon.

2

Hydrocarbon	Molecular formula	Boiling point (°C)
Methane	CH_4	–162
Ethane	C_2H_6	–89
Propane	C_3H_8	–42
Butane	C_4H_{10}	–0.5
Pentane	C_5H_{12}	36
Hexane	C_6H_{14}	69
Heptane	C_7H_{16}	98
Octane	C_8H_{18}	126

You may notice that there is a trend among the molecular formulae and boiling points of the hydrocarbons shown in Table 26.1. These hydrocarbons are members of the homologous series alkanes. So, what is a homologous series? Is there any trend among properties of members in the same homologous series? Can we predict the properties of an unfamiliar member of a homologous series from those of known members, in a way similar to working with elements of the same group of the periodic table?

26.2 Chemistry of carbon compounds

Simple carbon-containing compounds such as oxides of carbon, carbonates, hydrogencarbonates, carbides, cyanides are regarded as inorganic compounds.

Hydrocarbons and their derivatives are the focus of the branch of chemistry known as chemistry of carbon compounds. These substances are also called 'organic compounds' because early chemists thought that such compounds were made by living things (the word 'organic' means living). Later this was found to be untrue. At present, the name 'carbon compound' is applied to all carbon-containing compounds.

Carbon — a unique element

Carbon forms many more compounds than other elements do.

There are millions of different carbon compounds. The ability of carbon to form so many compounds lies in its unique structural and bonding properties.

- Carbon forms strong covalent bonds with itself to give chains and rings of its atoms (Fig. 26.1).

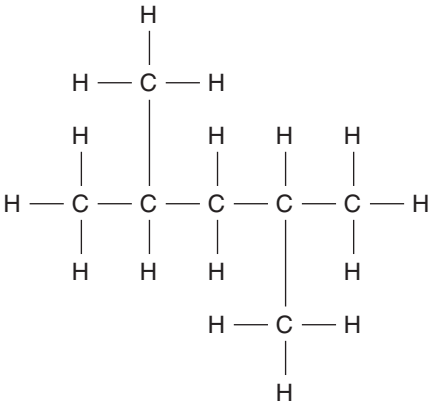


Fig. 26.1a A hydrocarbon with a branched chain

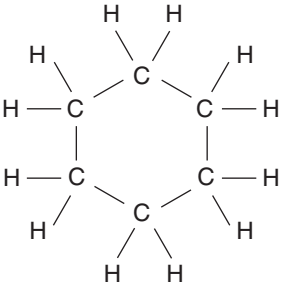


Fig. 26.1b A hydrocarbon with a ring structure

- Each carbon atom can form four covalent bonds, so the chains may be straight or branched, and can have other atoms or groups attached to them.

CE 2007 Paper 2 Q30

- The four covalent bonds of a carbon atom are actually arranged in a tetrahedral shape (Fig. 26.2).

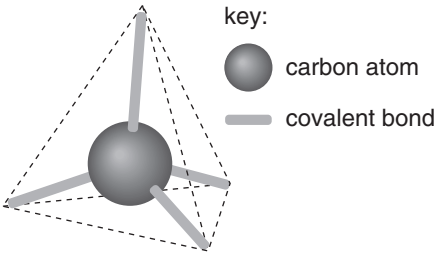


Fig. 26.2 The four covalent bonds of a carbon atom are arranged in a tetrahedral shape

However, for convenience sake, the bonds are written like this:



- One carbon atom can also form double and triple covalent bonds with another carbon atom (Table 26.2).

CE 2009 Paper 2 Q3

A carbon atom can also form double and triple covalent bonds with other atoms, e.g.

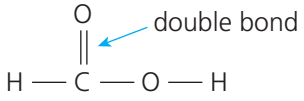
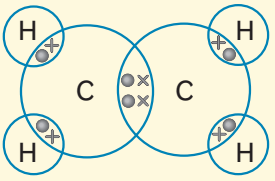
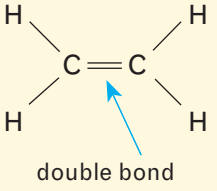
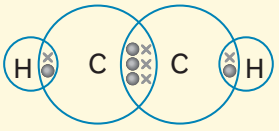
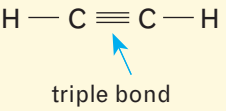


Table 26.2

Compounds containing a double bond or triple bond between two carbon atoms

Compound	Electron diagram	Structure
Ethene		
Ethyne		

26.3 Representing molecules of carbon compounds

We have already come across empirical formulae. An empirical formula of a covalent compound gives the simplest whole number ratio of atoms of each element in the compound. For example, the empirical formula for glucose is CH_2O. This tells us that there are two hydrogen atoms to each carbon atom and each oxygen atom. The molecular formula for glucose is $C_6H_{12}O_6$ and tells us how many of each type of atom there is in a molecule. However, the molecular formula does not show how the atoms are arranged in the molecule.

Structural formulae of butane and methylpropane

CE 2005 Paper 1 Q5(c)

Consider the following two compounds: butane and methylpropane.

Butane and methylpropane have the same molecular formula C_4H_{10}, but they are different compounds. Atoms of their molecules link together in different ways (Fig. 26.3).

Compounds with the same molecular formula but with their atoms arranged in different ways are isomers. We will discuss isomers in Topic 8 Chemistry of Carbon Compounds.

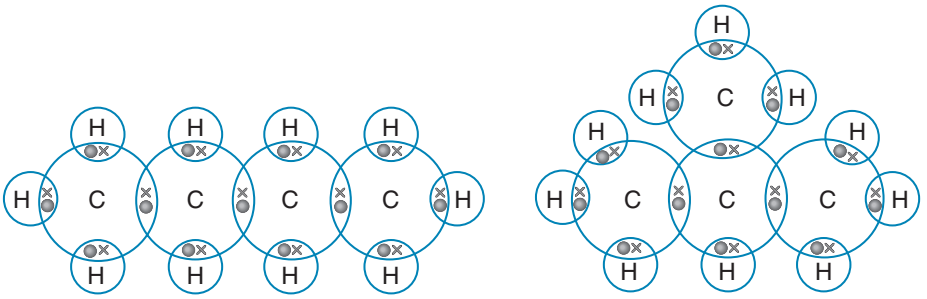


Fig. 26.3 Different compounds with the same molecular formula (only electrons in the outermost shells are shown here)

N1 Refer to page T20.

N2 Refer to page T20.

The **structural formula** of a compound shows how the atoms N1 are linked together. The two compounds above can be represented by the structural formulae shown in Fig. 26.4. N2

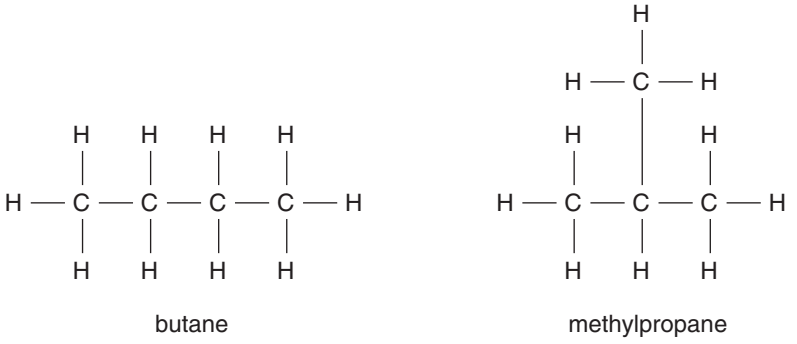


Fig. 26.4 Structural formulae of butane and methylpropane

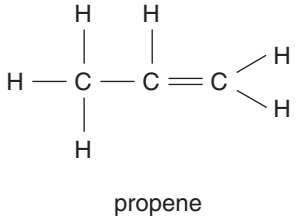


Fig. 26.5 Structural formula of propene

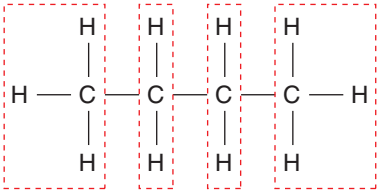
Structural formula of propene

Propene (C_3H_6) has a carbon-carbon double bond. Two lines are drawn between the carbon atoms to show the pairs of shared electrons (Fig. 26.5).

structural formula 結構式

Condensed structural formulae

Another way for writing the structural formula of a compound is the **condensed structural formula**. This method emphasizes groups of atoms in a compound. For example, the structural formula of butane may be written as shown in Fig. 26.6.



or $CH_3CH_2CH_2CH_3$
(condensed structural formula)

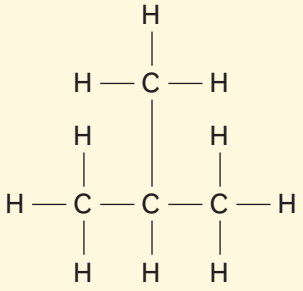
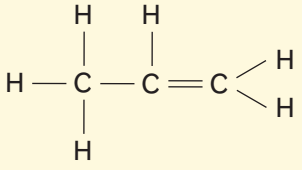
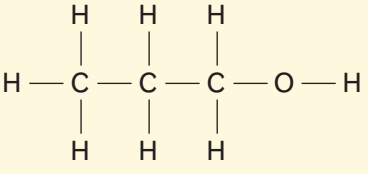
or $CH_3(CH_2)_2CH_3$
(condensed structural formula)

Fig. 26.6 The condensed structural formula of butane

In condensed structural formulae, we omit single bonds (except those joining anything other than hydrogen atoms to the main carbon chain). However, we must write carbon-carbon multiple bonds (that is C=C or C≡C). Table 26.3 shows more examples.

Table 26.3

Some examples of structural formulae and condensed structural formulae

Structural formula	Condensed structural formula
	CH_3CHCH_3 or $CH_3CH(CH_3)CH_3$
	$CH_3CH{=}CH_2$
	$CH_3CH_2CH_2OH$

condensed structural formula 簡明結構式

Skeletal formulae

Formulae that only show how the carbon atoms are bonded are called **skeletal formulae**. In these formulae, all the carbon and hydrogen atoms are omitted and only heteroatoms (e.g. oxygen or nitrogen) are shown explicitly.

Skeletal formulae depict the carbon skeleton of a molecule using a *zig-zag line*. Each intersection or end of line represents a carbon atom with the appropriate number of hydrogen atoms. Fig. 26.7 shows the skeletal formula of butane.

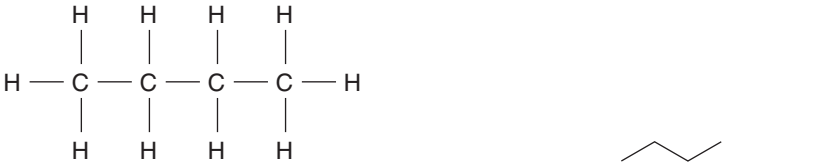
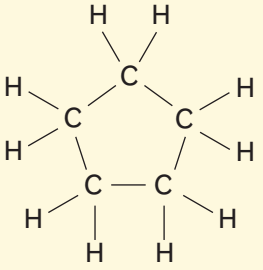
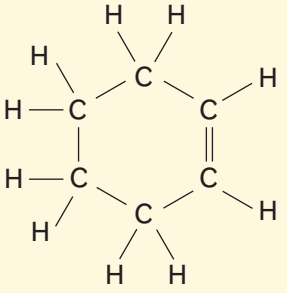


Fig. 26.7 Skeletal formula of butane

The skeletal formulae we encounter most are those of ring compounds. Table 26.4 shows two examples.

Table 26.4

Two examples of the structural and skeletal formulae of ring compounds

Structural formula	Skeletal formula
	
	

skeletal formula 骨架式　　zig-zag line 鋸齒形線

The skeletal method is particularly useful when writing the formulae of large molecules, such as cholesterol (Fig. 26.8).

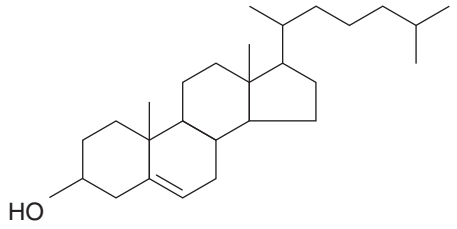
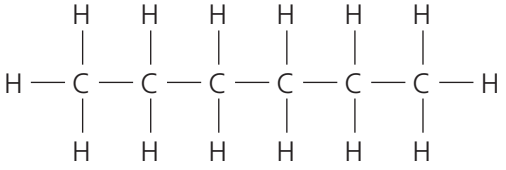


Fig. 26.8 Skeletal formula of cholesterol

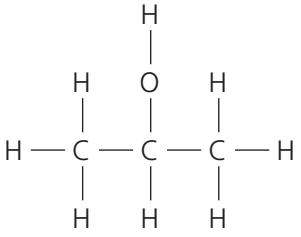
Practice 26.1

1 Write the condensed structural formulae of the following compounds.

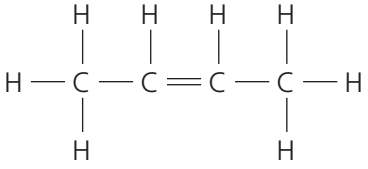
a)



b)

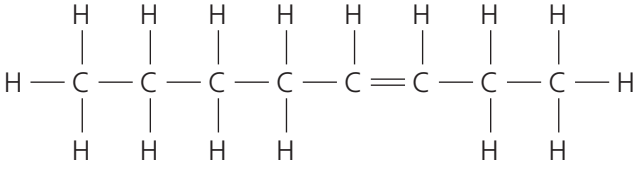


c)

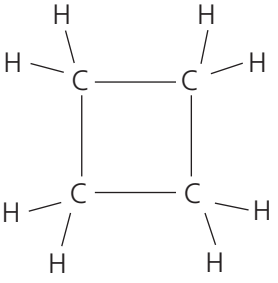


2 Write the skeletal formulae of the following compounds.

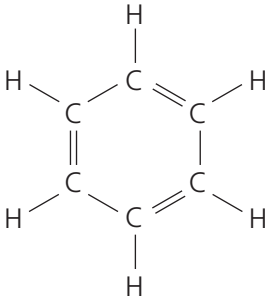
a)



b)



c)



1 a) $CH_3CH_2CH_2CH_2CH_2CH_3$ or $CH_3(CH_2)_4CH_3$

b) $CH_3\underset{\displaystyle OH}{\underset{|}{C}}HCH_3$ or $CH_3CH(OH)CH_3$

c) $CH_3CH{=}CHCH_3$

2 a)



b)



c)



26.4 Molecular models

Animation
Molecular model

The molecular structures of carbon compounds are actually three-dimensional. It is useful to use **molecular models** to show the exact molecular structures and shapes of carbon compounds. The simplest models are **ball-and-stick models** (Fig. 26.9a). To get a more accurate idea of how close together different atoms are in a compound, we use **space-filling models** (Fig. 26.9b).

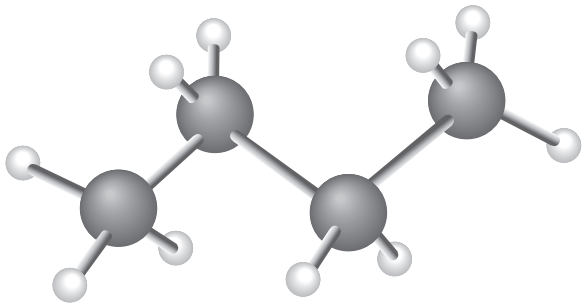


Fig. 26.9a A ball-and-stick model of butane

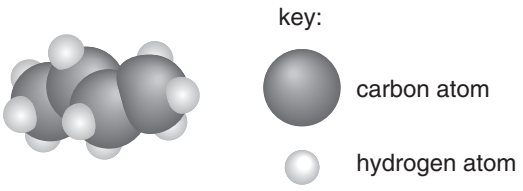
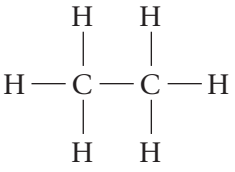


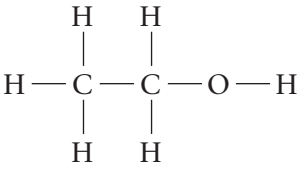
Fig. 26.9b A space-filling model of butane

26.5 Functional groups

Consider two compounds — ethane and ethanol.



ethane



ethanol

Ethane and ethanol have similar molecular structures except that one hydrogen atom in ethane is replaced by a hydroxyl group –O–H in ethanol. However, they have very different properties (Table 26.5).

Table 26.5

Some properties of ethane and ethanol

Property	Ethane	Ethanol
State at room temperature and pressure	gas	liquid
Reaction with sodium	no reaction	reacts to give hydrogen gas

molecular model 分子模型　　ball-and-stick model 球棒模型　　space-filling model 填空模型

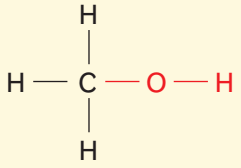
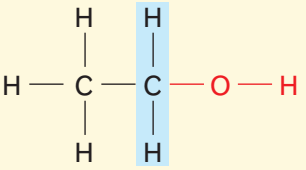
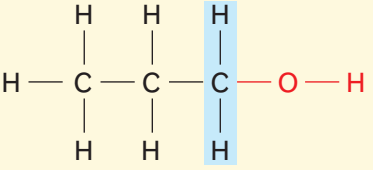
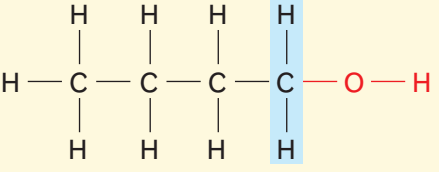
Clearly, the hydroxyl group –O–H in ethanol has a great effect in modifying the properties of the ethane skeleton. The hydroxyl group –O–H in ethanol is an example of a **functional group**.

✓ A functional group is an atom, or a group of atoms, which determines most of the properties of a compound.

26.6 Homologous series

A family of compounds containing the same functional group is called a **homologous series**. Methanol, ethanol, propan-1-ol and butan-1-ol are the first four members of the homologous series of alkanols (Table 26.6). Their structural formulae show that each alkanol differs from the next by a $-CH_2-$ unit.

Table 26.6
The first four members of the homologous series of alkanols

Name	Molecular formula	Structural formula
Methanol	CH_3OH	
Ethanol	C_2H_5OH	
Propan-1-ol	C_3H_7OH	
Butan-1-ol	C_4H_9OH	

functional group 官能基　　homologous series 同系列

A functional group reacts in nearly the same way, whatever the size of the hydrocarbon portion. So, for example, if we learn the reactions of one alkanol, such as ethanol, we can apply this knowledge to any other alkanols.

However, the size of the hydrocarbon portion affects physical properties, like melting point, boiling point and water solubility. Thus, members of a homologous series show a steady gradation in physical properties as the length of the carbon chain in molecules increases.

Members in a homologous series show the following characteristics:

1 They have the same **general formula**. For example, the general formula of alkanols is $C_nH_{2n+1}OH$, where n = 1, 2, 3, 4, etc.

2 Each member in a homologous series differs from the next by a $-CH_2-$ unit.

3 They show a gradual change in physical properties.

4 They show similar chemical properties.

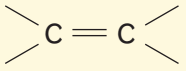
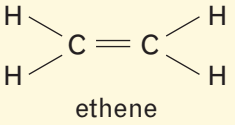
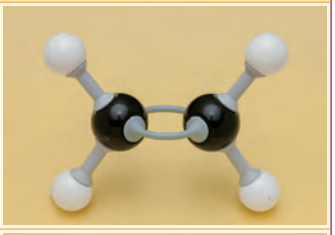
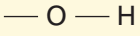
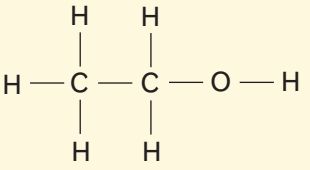
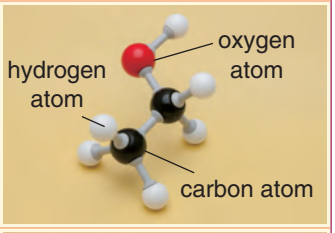
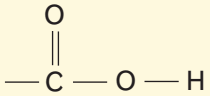
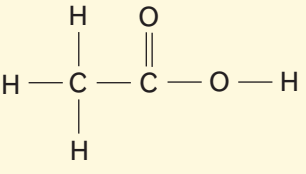
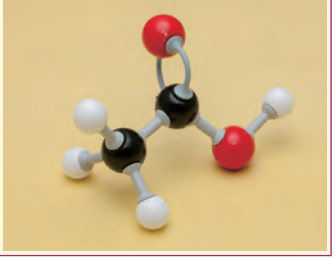
26.1
Building molecular models of alkanols and alkanoic acids.

Table 26.7 shows some common functional groups and homologous series.

CE 2005 Paper 1 Q5(a)
CE 2007 Paper 2 Q4

Table 26.7

Some common functional groups

Functional group		Homologous series		Example	
Formula	Name	General formula	Name		
	—	C_nH_{2n}	alkene	 ethene	
	hydroxyl group	$C_nH_{2n+1}OH$	alkanol	 ethanol	
	carboxyl group	$C_nH_{2n+1}COOH$	alkanoic acid	 ethanoic acid	

general formula 通式

26.7 The alkanes

The major component of natural gas is methane. Each methane molecule contains one carbon atom and four hydrogen atoms (Fig. 26.10). Methane is the simplest member of the homologous series of alkanes.

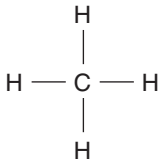


Fig. 26.10 Structural formula of methane

The alkanes are a family of **saturated hydrocarbons** — all the carbon atoms are connected to each other by single bonds.

The general formula of members of the alkane series is C_nH_{2n+2}, where n = 1, 2, 3, 4, etc. It is possible to have alkanes with straight or branched chains (Fig. 26.11).

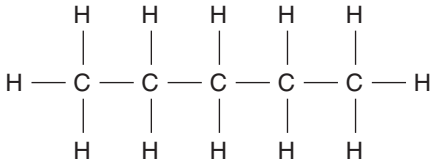


Fig. 26.11a A straight-chain alkane

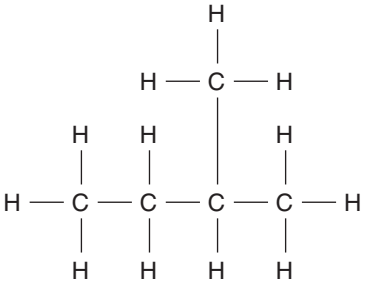


Fig. 26.11b A branched-chain alkane

saturated hydrocarbon 飽和烴

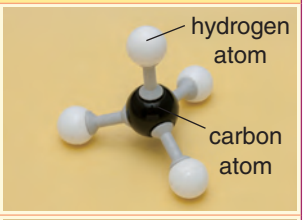
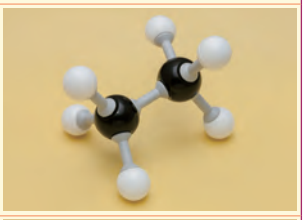
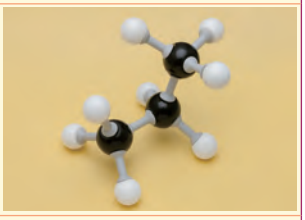
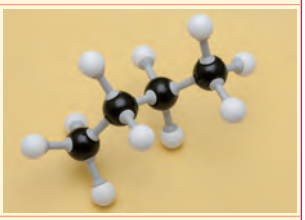
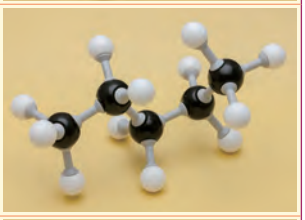
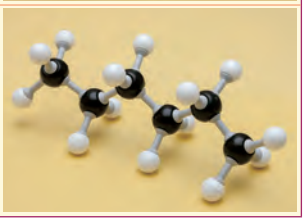
26.2

Building molecular models of alkanes.

Table 26.8 shows the condensed structural formulae and molecular models of six simple straight-chain alkanes.

Table 26.8

Condensed structural formulae and molecular models of six simple straight-chain alkanes

Name	Molecular formula	Condensed structural formula	Molecular model
Methane	CH_4	CH_4	
Ethane	C_2H_6	CH_3CH_3	
Propane	C_3H_8	$CH_3CH_2CH_3$	
Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$	
Pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$	
Hexane	C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$	

26.8 Naming straight-chain and branched-chain alkanes

A systematic way of naming carbon compounds is used throughout the world. It is known as the IUPAC (*International Union of Pure and Applied Chemistry*) system.

Reference Website

Refer to the following website for tutorials and exercises on the nomenclature of organic compounds:
http://cd1.edb.hkedcity.net/cd/science/chemistry/resource/naming/t4_main.htm
(accessed August 2014)

Straight-chain alkanes

To name straight-chain alkanes, use a **prefix** to show how many carbon atoms are in the straight chain, followed by the **suffix** -ane. Table 26.9 shows the names and prefixes for the first eight alkanes.

Table 26.9

Prefixes and names for the first eight alkanes

Structural formula of alkane	Number of carbon atoms per molecule	Prefix	Name of alkane
CH_4	1	meth-	methane
CH_3CH_3	2	eth-	ethane
$CH_3CH_2CH_3$	3	prop-	propane
$CH_3CH_2CH_2CH_3$	4	but-	butane
$CH_3CH_2CH_2CH_2CH_3$	5	pent-	pentane
$CH_3CH_2CH_2CH_2CH_2CH_3$	6	hex-	hexane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	7	hept-	heptane
$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	8	oct-	octane

Branched-chain alkanes

To name branched-chain alkanes, consider them as straight-chain alkanes with hydrogen atoms replaced by alkyl groups◂.

Alkyl groups are often represented by the symbol R.

Alkyl groups are derived from alkanes by the removal of a hydrogen atom. Name an alkyl group by replacing the suffix -ane of the parent alkane by -yl.

International Union of Pure and Applied Chemistry 國際純粹與應用化學聯合會 prefix 詞首 suffix 詞尾

Table 26.10 gives the names and the formulae of the first four alkyl groups.

Table 26.10

Names and formulae of the first four alkyl groups

Name	Formula
Methyl	$CH_3–$
Ethyl	$C_2H_5–$
Propyl	$C_3H_7–$
Butyl	$C_4H_9–$

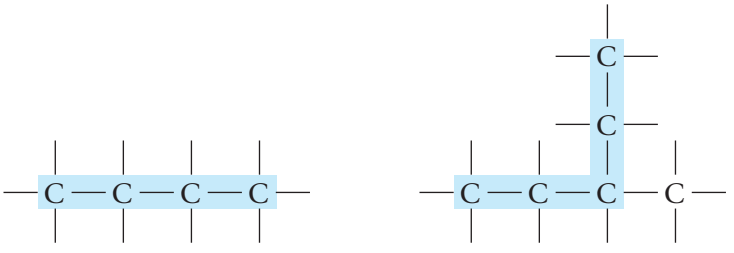
The IUPAC name for a branched-chain alkane consists of two parts:



The IUPAC rules for naming branched-chain alkanes are as follows:

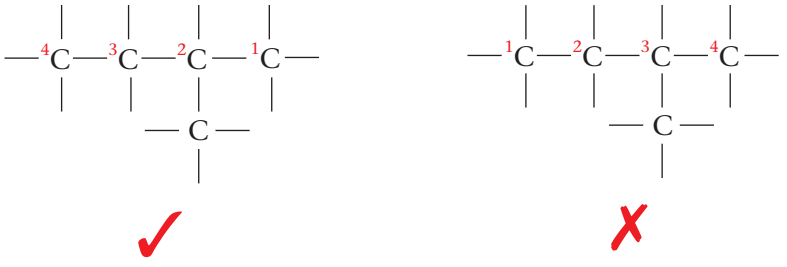
1 Locate the longest continuous chain of carbon atoms. This is the parent chain. Use the IUPAC system to name the parent chain. Notice that the longest continuous carbon chain may not appear in a straight line.

Examples:



2 Number the parent chain, starting from the end nearest to the alkyl group(s).

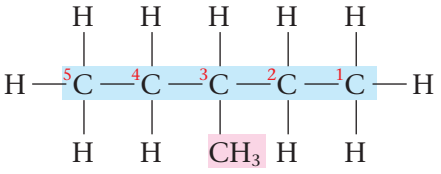
Example:



✓ ✗

3 Name each alkyl group. Assign a number to each alkyl group, matching the number of the carbon atom where it attaches to the parent chain. Place the number in front of the name of each alkyl group, followed by a hyphen.

Example:

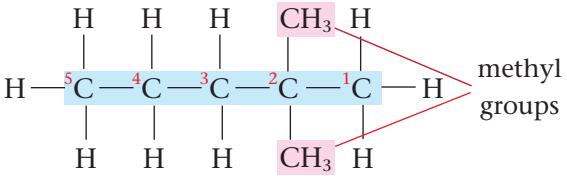


3-methylpentane

4 If two or more identical alkyl groups are attached to the parent chain, specify this by numerical prefixes.

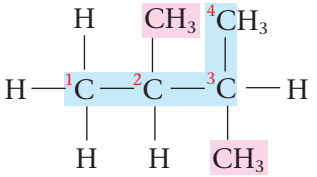
Number of identical alkyl groups	Numerical prefix
2	di-
3	tri-
4	tetra-

Examples:



methyl groups

2,2-dimethylpentane



2,3-dimethylbutane

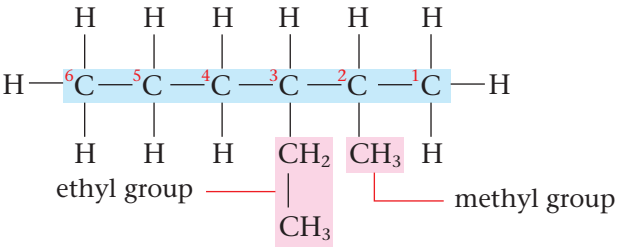
Separate two numbers by a comma ‘,’.

ASL 2008 Paper 2 Q2(a)(i)
AL 2008 Paper 2 Q5(a)(i)

CE 2005 Paper 2 Q1
ASL 2006 Q11(a)(i)

5 If two or more different alkyl groups are attached to the parent chain, list them in alphabetical order (for example, ethyl before methyl).

Example:

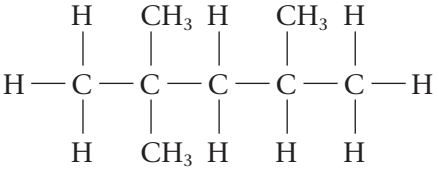


3-ethyl-2-methylhexane

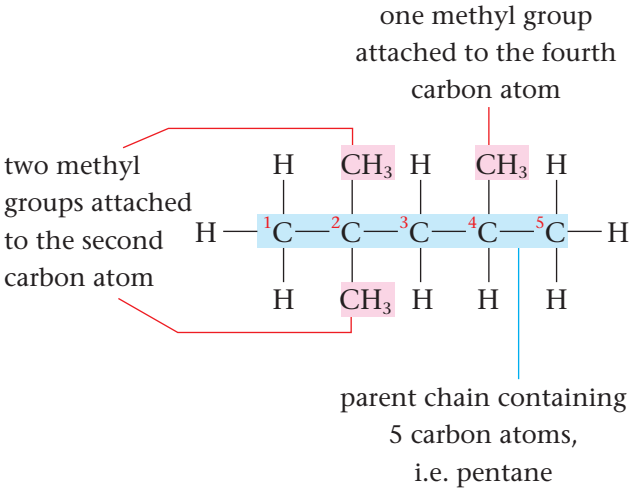
Use a hyphen '-' between a number and a letter. Notice that 'ethyl' is listed before 'methyl' although the methyl group is attached to the second carbon atom and the ethyl group to the third atom.

Example 26.1

Q Name the following compound by the IUPAC system.



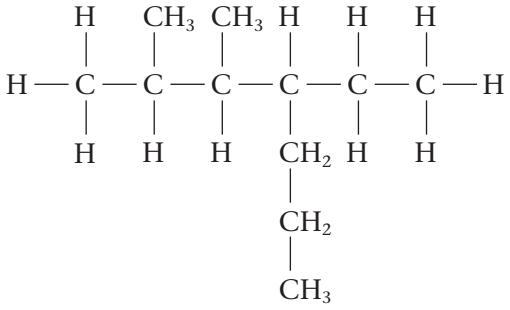
A



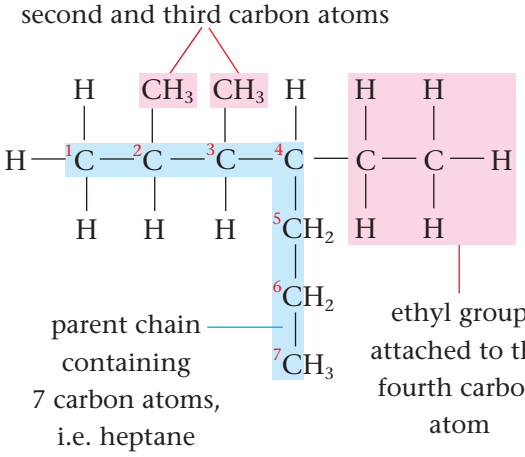
The IUPAC name of the compound is 2,2,4-trimethylpentane.

Example 26.2

Q Name the following compound by the IUPAC system.



A two methyl groups attached to the second and third carbon atoms



The IUPAC name of the compound is 4-ethyl-2,3-dimethylheptane.

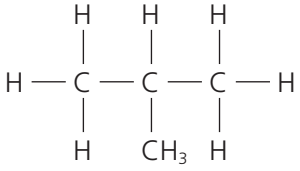
Notice that when deciding alphabetical order of the prefixes such as 'ethyl' and 'methyl', ignore the numerical prefixes, such as 'di'.

Practice 26.2

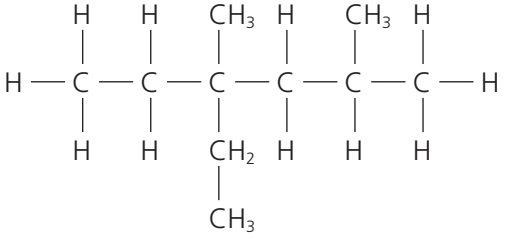
Refer to page T21.

1 Name the following compounds by the IUPAC system.

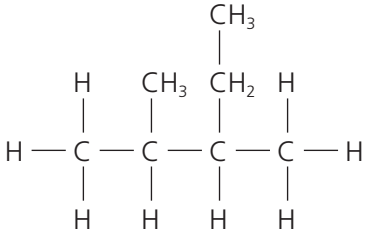
a)



c)



b)



d) $CH_3CH(CH_3)CH_2CH_3$

2 Write the structural formulae of the following compounds.

a) 2,2-dimethylbutane

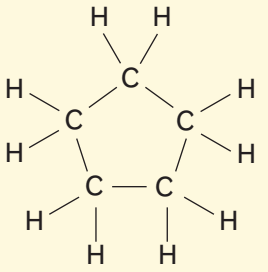
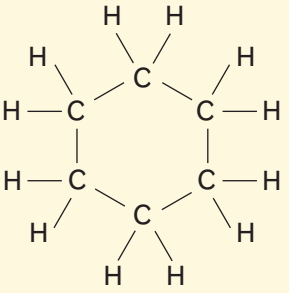
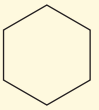
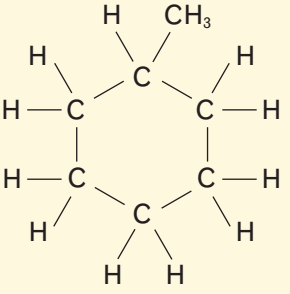
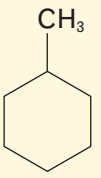
b) 3-ethyl-3-methylhexane

Cycloalkanes

The alkanes considered so far all have open-chain molecules. It is also possible for alkane molecules to have ring structures. These compounds are cycloalkanes. The general formula is C_nH_{2n}, where n = 3, 4, 5, 6, etc. Cycloalkanes are named by using the prefix cyclo-. Some cycloalkanes are shown in Table 26.11.

Table 26.11

Some examples of cycloalkanes

	Structural formula	Skeletal formula
Cyclopentane (C_5H_{10})		
Cyclohexane (C_6H_{12})		
Methylcyclohexane (C_7H_{14})		

26.9 Naming alkenes

Like the alkanes, the alkenes are hydrocarbons, but they are unsaturated. An **unsaturated hydrocarbon** has one or more double (C=C) or triple bonds (C≡C) between the carbon atoms (Fig. 26.12).

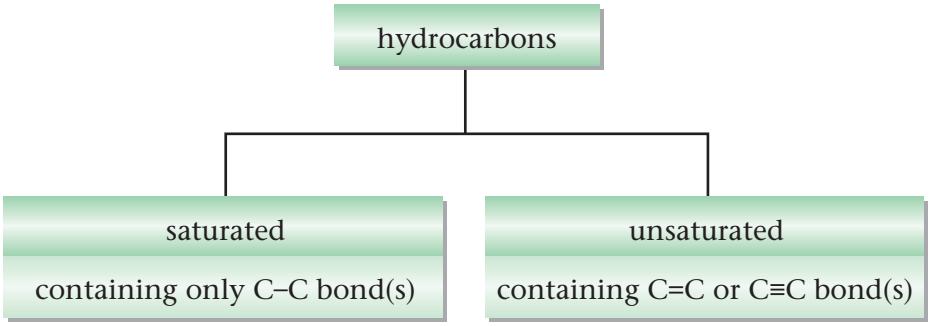


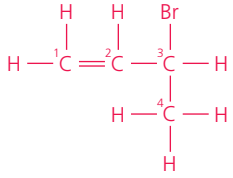
Fig. 26.12 Hydrocarbons can be classified as saturated and unsaturated

All alkenes contain at least one carbon-carbon double bond. Alkenes with one double bond per molecule have the general formula C_nH_{2n}. Name the alkenes with the rules as if for alkanes, but using the suffix -ene instead of -ane.

The IUPAC rules for naming alkenes are as follows:

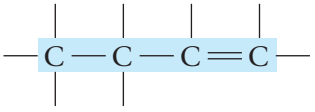
1 Locate the longest continuous carbon chain that contains the double bond. Determine the parent name by changing the ending of the name of the alkane of identical length from -ane to -ene.

N3

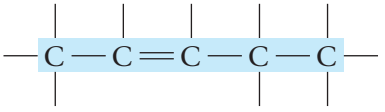


The IUPAC name of the above compound is 3-bromobut-1-ene.

Examples:



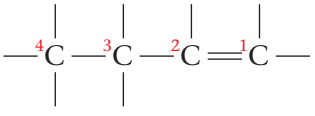
butene parent chain N3



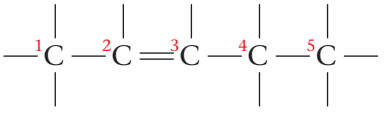
pentene parent chain

2 Number the parent chain, starting from the end nearest to the double bond. Indicate the position of the double bond by the number of the first carbon atom of the double bond. Place the number in front of the suffix -ene.

Examples:



but-1-ene
(not but-3-ene)

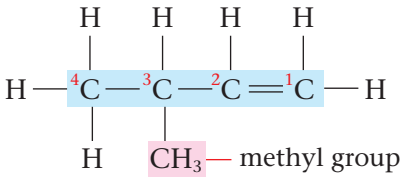


pent-2-ene
(not pent-3-ene)

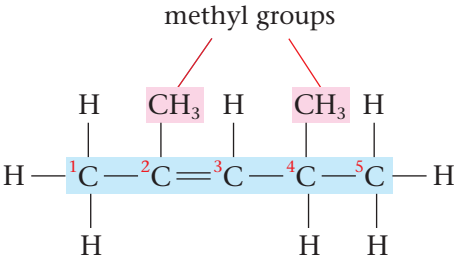
unsaturated hydrocarbon 不飽和烴

3 Name each alkyl group. Assign a number to each alkyl group, matching the number of the carbon atom where it attaches to the parent chain.

Examples:



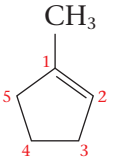
3-methylbut-1-ene



2,4-dimethylpent-2-ene

4 For a cycloalkene with attached alkyl group(s), number the carbon atoms of the double bond the 1 and 2 positions. Give the alkyl group(s) the lower number(s) as well.

Example:



1-methylcyclopentene

It is not necessary to specify the position of the carbon-carbon double bond since it is always at the first and second carbon atoms.

26.3

Building molecular models of alkenes.

Table 26.12 shows the condensed structural formulae and names of six simple alkenes.

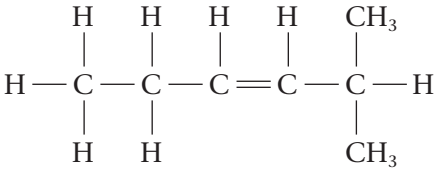
Table 26.12

Condensed structural formulae and names of six simple alkenes and their corresponding alkanes

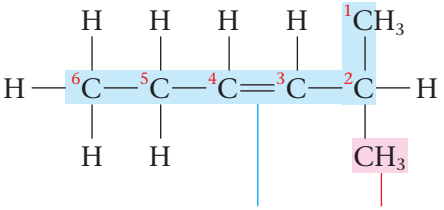
Condensed structural formula of alkene	Name	Corresponding alkane
$CH_2=CH_2$	ethene	ethane (CH_3CH_3)
$CH_3CH=CH_2$	propene	propane ($CH_3CH_2CH_3$)
$CH_3CH_2CH=CH_2$	but-1-ene	butane ($CH_3CH_2CH_2CH_3$)
$CH_3CH=CHCH_3$	but-2-ene	butane ($CH_3CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH=CH_2$	pent-1-ene	pentane ($CH_3CH_2CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH_2CH=CH_2$	hex-1-ene	hexane ($CH_3CH_2CH_2CH_2CH_2CH_3$)

Example 26.3

Q Name the following compound by the IUPAC system.



A



parent chain containing 6 carbon atoms, with the double bond between the third and fourth carbon atoms, i.e. hex-3-ene

methyl group attached to the second carbon atom

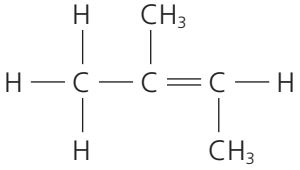
The name of the compound is 2-methylhex-3-ene.

Practice 26.3

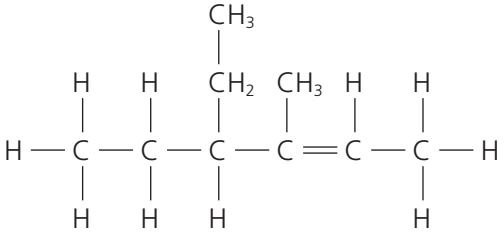
Refer to page T22.

1 Name the following compounds by the IUPAC system.

a)

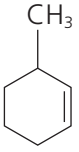


b)



c) $(CH_3)_2CHCH_2CH=CH_2$

d)



2 Write the structural formulae of the following compounds.

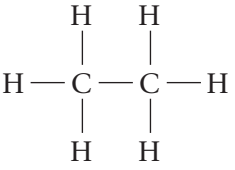
a) 2,3-dimethylbut-1-ene

b) 3-ethylpent-2-ene

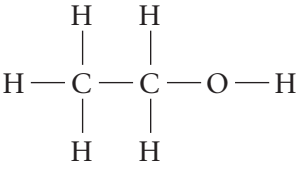
26.10 Naming alkanols

Alkanols have the general formula $C_nH_{2n+1}OH$, where n = 1, 2, 3, 4, etc. The C_nH_{2n+1}– is an alkyl group while the **hydroxyl group** –O–H is the functional group.

Alkanols are named by the IUPAC system as derivatives of the parent alkane. The last letter 'e' in the name of the corresponding alkane is replaced by the suffix -ol. For example, ethanol is derived from ethane.



ethane



ethanol

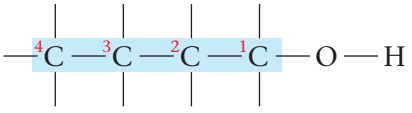
The IUPAC rules for naming alkanols are as follows:

1 Locate the longest continuous carbon chain to which the –O–H group is attached. Change the name of the corresponding alkane by replacing the last letter 'e' by the suffix -ol.

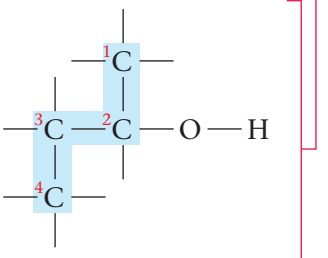
2 Number the parent chain, starting from the end nearest to the –O–H group. Indicate the position of the –O–H group by a number in front of the suffix -ol.

Examples:

CE 2010 Paper 2 Q12



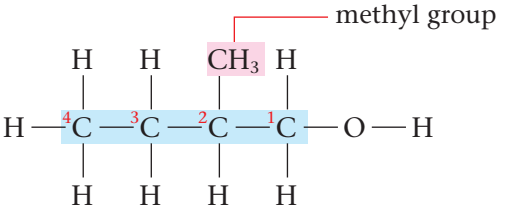
butan-1-ol



butan-2-ol N4

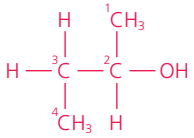
3 Name each alkyl group. Assign a number to each alkyl group, matching the number of the carbon atom where it attaches to the parent chain.

Example:

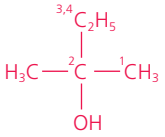


2-methylbutan-1-ol

N4



The IUPAC name of the above compound is butan-2-ol.



The IUPAC name of the above compound is 2-methylbutan-2-ol.

alkanol 烷醇 hydroxyl group 羥基

Table 26.13 shows the condensed structural formulae and names of six simple alkanols.

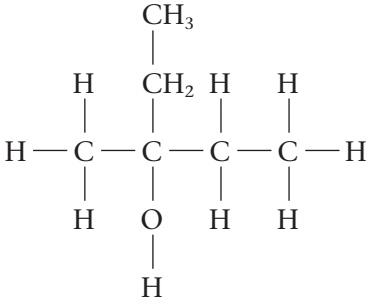
Table 26.13

Condensed structural formulae and names of six simple alkanols and their corresponding alkanes

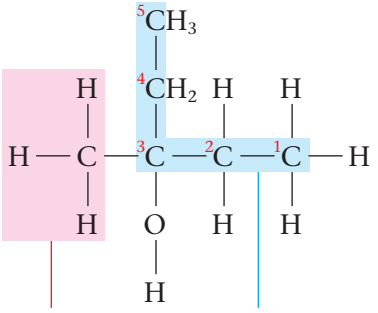
Condensed structural formula of alkanol	Name	Corresponding alkane
CH_3OH	methanol	methane (CH_4)
CH_3CH_2OH	ethanol	ethane (CH_3CH_3)
$CH_3CH_2CH_2OH$	propan-1-ol	propane ($CH_3CH_2CH_3$)
$CH_3CH_2CH_2CH_2OH$	butan-1-ol	butane ($CH_3CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH_2CH_2OH$	pentan-1-ol	pentane ($CH_3CH_2CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH_2CH_2CH_2OH$	hexan-1-ol	hexane ($CH_3CH_2CH_2CH_2CH_2CH_3$)

Example 26.4

Q Name the following compound by the IUPAC system.



A



methyl group attached to the third carbon atom

parent chain containing 5 carbon atoms, with the –O–H group attached to the third carbon atom, i.e. pentan-3-ol

The IUPAC name of the compound is 3-methylpentan-3-ol.

26.11 Naming alkanoic acids

Alkanoic acids have the general formula RCOOH, where R is an alkyl group or a hydrogen atom. The **carboxyl group**  is the functional group.

Name an alkanoic acid by replacing the last letter 'e' in the name of the corresponding alkane by a suffix -oic acid. The carbon atom of the carboxyl group is assigned number 1. Table 26.14 shows the condensed structural formulae and names of six simple alkanoic acids.

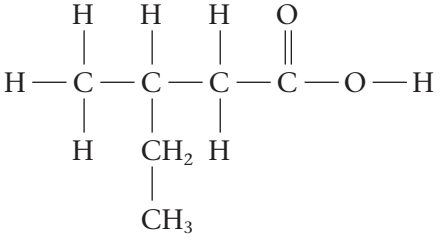
Table 26.14

Condensed structural formulae and names of six simple alkanoic acids and their corresponding alkanes

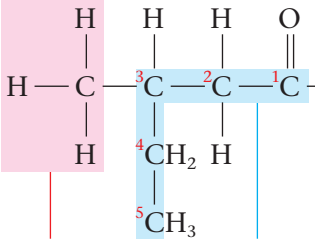
Condensed structural formula of alkanoic acid	Name	Corresponding alkane
HCOOH	methanoic acid	methane (CH_4)
CH_3COOH	ethanoic acid	ethane (CH_3CH_3)
CH_3CH_2COOH	propanoic acid	propane ($CH_3CH_2CH_3$)
$CH_3CH_2CH_2COOH$	butanoic acid	butane ($CH_3CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH_2COOH$	pentanoic acid	pentane ($CH_3CH_2CH_2CH_2CH_3$)
$CH_3CH_2CH_2CH_2CH_2COOH$	hexanoic acid	hexane ($CH_3CH_2CH_2CH_2CH_2CH_3$)

Example 26.5

Q Name the following compound by the IUPAC system.



A



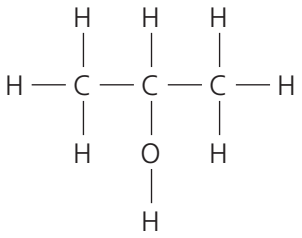
The IUPAC name of the compound is 3-methylpentanoic acid.

alkanoic acid 烷酸　　carboxyl group 羧基

Practice 26.4 Refer to page T23.

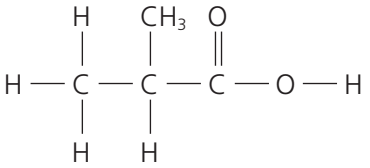
1 Name the following compounds by the IUPAC system.

a)



b) $CH_3CH_2CH_2CH(OH)CH_3$

c)



2 Write the structural formulae of the following compounds.

a) 3-methylbutan-1-ol

b) 2-methylbutanoic acid

26.12 Physical properties of alkanes

Melting and boiling points

Table 26.15 shows the melting and boiling points of several straight-chain alkanes.

Table 26.15

The melting and boiling points of several straight-chain alkanes

Name	Molecular formula	Melting point (°C)	Boiling point (°C)	State at room temperature and pressure
Methane	CH_4	–182	–162	gas
Ethane	C_2H_6	–183	–89	
Propane	C_3H_8	–190	–42	
Butane	C_4H_{10}	–138	–0.5	
Pentane	C_5H_{12}	–130	36	liquid
Hexane	C_6H_{14}	–95	69	
Heptane	C_7H_{16}	–91	98	
Octane	C_8H_{18}	–57	126	
⋮	⋮	⋮	⋮	
Heptadecane	$C_{17}H_{36}$	22	302	
Octadecane	$C_{18}H_{38}$	28	316	solid
Nonadecane	$C_{19}H_{40}$	32	330	

CE 2008 Paper 2 Q29

The only intermolecular forces between alkane molecules are van der Waals' forces. The relative molecular mass of an alkane increases as the number of carbon atoms per molecule increases. The molecular size and the number of electrons per molecule also increase. Hence the van der Waals' forces between the molecules increase. More heat is needed to separate the molecules during melting and boiling. Thus, the melting points and boiling points of alkanes rise as the number of carbon atoms in the alkane molecules increases (Fig. 26.13).

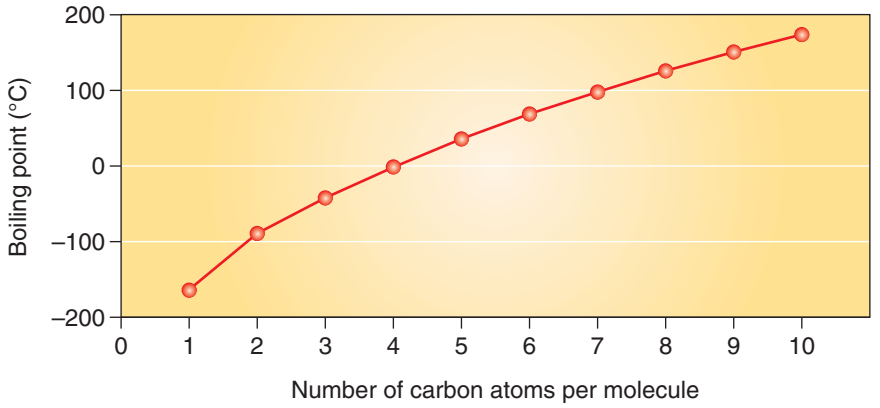


Fig. 26.13 The graph shows the gradual change of boiling points of straight-chain alkanes as the number of carbon atoms per molecule increases

Solubility

Alkanes are insoluble in water. This is because water molecules are held together by strong hydrogen bonds. The weak attractive forces between alkane and water molecules are not strong enough to overcome the hydrogen bonds between water molecules. However, alkanes do mix with other relatively non-polar liquids.

Density

Liquid alkanes are less dense than water. Their densities increase with the relative molecular masses.

Example 26.6

Q Explain why heptadecane ($C_{17}H_{36}$) found in diesel oil is more viscous than heptane (C_7H_{16}) found in petrol.

A A heptadecane molecule has a larger relative molecular mass / a large molecular size (or a greater number of electrons) than a heptane molecule. Thus, heptadecane has stronger van der Waals' forces than heptane.

Also, heptadecane molecules in the form of long chains can become entangled. Hence the relative motion between the molecules is hindered.

Thus, heptadecane is more viscous than heptane.

Have you mastered?

Key terms

structural formula 結構式 38 condensed structural formula 簡明結構式 39

skeletal formula 骨架式 40 molecular model 分子模型 42 ball-and-stick model 球棒模型 42

space-filling model 填空模型 42 functional group 官能基 43 homologous series 同系列 43

general formula 通式 44 saturated hydrocarbon 飽和烴 45 prefix 詞首 47

suffix 詞尾 47 parent chain 母鏈 48 alkene 烯 53 unsaturated hydrocarbon 不飽和烴 53

alkanol 烷醇 56 hydroxyl group 羥基 56 alkanoic acid 烷酸 58 carboxyl group 羧基 58

Checklist

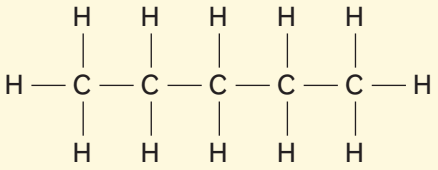
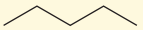
After studying this unit, you should be able to

- ☐ explain the large number and diversity of carbon compounds with reference to carbon's unique combination power and ability to form different bonds;
- ☐ explain the meaning of a homologous series;
- ☐ understand that members of a homologous series show a gradation in physical properties and similarity in chemical properties;
- ☐ write structural formulae of alkanes, alkenes, alkanols and alkanoic acids;
- ☐ write skeletal formulae of cycloalkanes;
- ☐ give systematic names of alkanes, cycloalkanes, alkenes, alkanols and alkanoic acids.

(Put a '✔' in the box if you have acquired the knowledge concerned.)

Summary

1 Carbon compounds can be represented by structural formulae, condensed structural formulae and skeletal formulae.

Structural formula	Condensed structural formula	Skeletal formula
	$CH_3(CH_2)_3CH_3$	

2 A functional group is an atom, or a group of atoms, which determines most of the properties of a compound.

3 A family of compounds containing the same functional group is called a homologous series.

Members in a homologous series show the following characteristics:

a) have the same general formula;

b) each member differs from the next by a $-CH_2-$ unit;

c) show a gradual change in physical properties;

d) show similar chemical properties.

4 Alkanes have the general formula C_nH_{2n+2}.

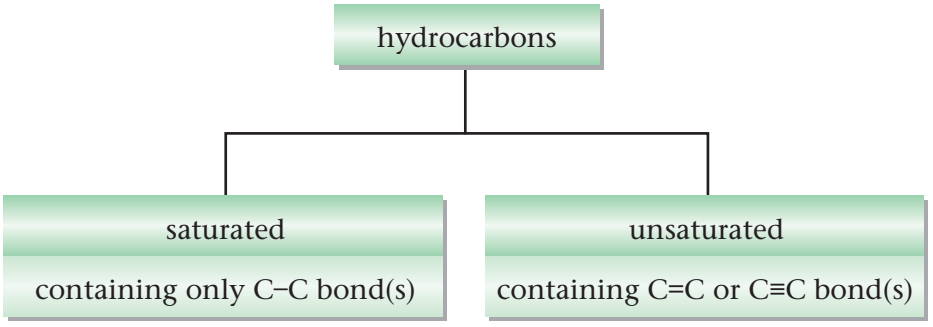
Name	Molecular formula	Condensed structural formula
Methane	CH_4	CH_4
Ethane	C_2H_6	CH_3CH_3
Propane	C_3H_8	$CH_3CH_2CH_3$
Butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$
Pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$
Hexane	C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$
Heptane	C_7H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$
Octane	C_8H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$

Have you mastered?

The IUPAC name for a branched-chain alkane consists of two parts:



5 Hydrocarbons can be classified as saturated and unsaturated.



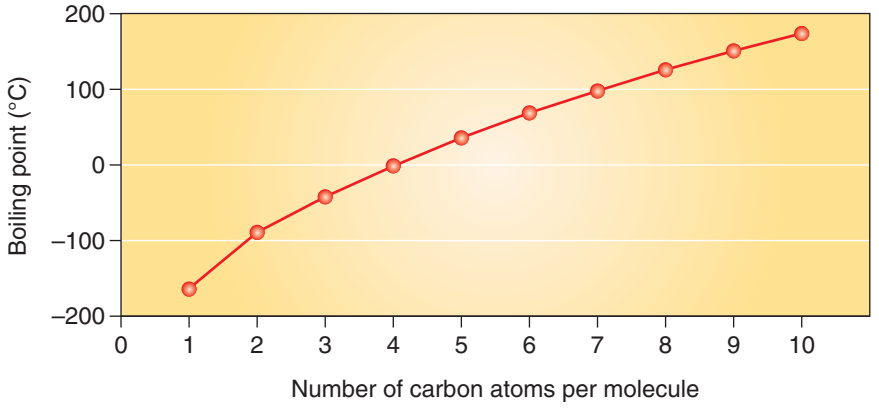
6 The following table summarizes the information for some common homologous series.

Homologous series	General formula	Functional group	Naming	Example(s)
Alkenes	C_nH_{2n}	$>C=C<$	name the compounds with the rules as if for alkanes, but using the suffix -ene instead of -ane	$CH_3CH_2CH=CH_2$ but-1-ene $CH_3CH=CHCH_3$ but-2-ene
Alkanols	$C_nH_{2n+1}OH$	—O—H	name the compound by replacing the last letter 'e' in the name of the corresponding alkane by the suffix -ol	$CH_3CH_2CH_2CH_2OH$ butan-1-ol $CH_3CH_2CH(OH)CH_3$ butan-2-ol
Alkanoic acids	$C_nH_{2n+1}COOH$	$-C(=O)-O-H$	name the compound by replacing the last letter 'e' in the name of the corresponding alkane by the suffix -oic acid; the carbon atom of the carboxyl group is assigned number 1	$CH_3CH_2CH_2COOH$ butanoic acid

Have you mastered?

7 a) The melting points and boiling points of alkanes rise as the number of carbon atoms in the alkane molecules increases.

Name	Molecular formula	Melting point (°C)	Boiling point (°C)	State at room temperature and pressure
Methane	CH_4	–182	–162	gas
Ethane	C_2H_6	–183	–89	
Propane	C_3H_8	–190	–42	
Butane	C_4H_{10}	–138	–0.5	
Pentane	C_5H_{12}	–130	36	liquid
Hexane	C_6H_{14}	–95	69	



b) Alkanes are insoluble in water.

c) Liquid alkanes are less dense than water.

Have you mastered?

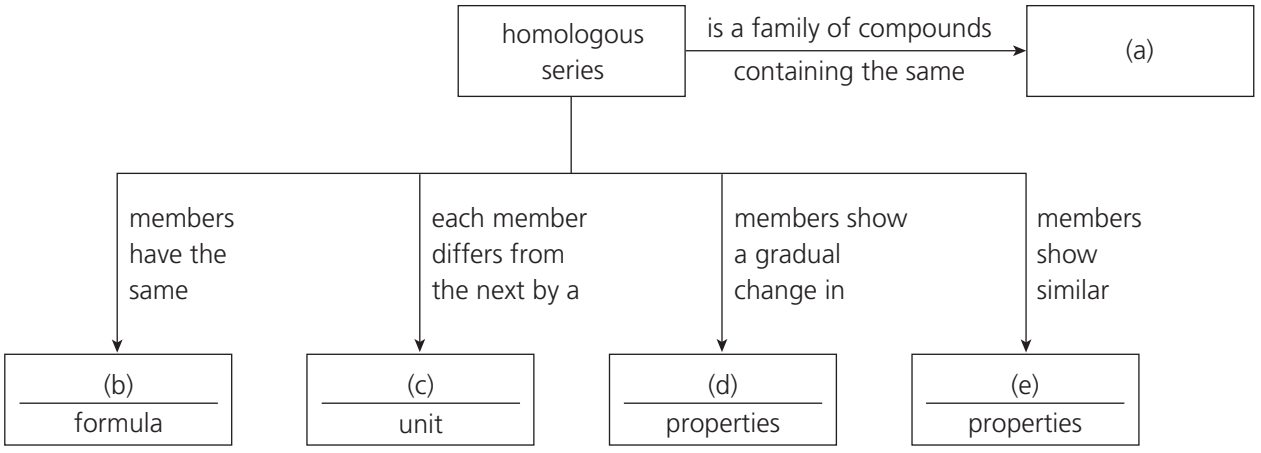
Unit Exercise

Note: The symbol ! indicates the level of difficulty of a question.

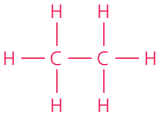
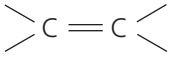
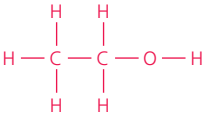
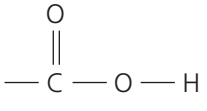
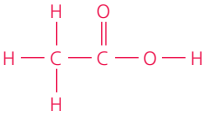
Part I Knowledge and understanding

a) functional group
b) general
c) $-CH_2-$
d) physical
e) chemical

1 Complete the following concept map.



2 Complete the following table.

Name of homologous series	Functional group	General formula	Example	
			Name	Structural formula
Alkane	—	C_nH_{2n+2}	ethane	
Alkene		C_nH_{2n}	ethene	
Alkanol	—O—H	$C_nH_{2n+1}OH$	ethanol	
Alkanoic acid		$C_nH_{2n+1}COOH$	ethanoic acid	

3 Complete the following table concerning some simple straight-chain alkanes.

Name	Molecular formula	Condensed structural formula
Methane	CH_4	CH_4
Ethane	C_2H_6	CH_3CH_3
Propane	C_3H_8	$CH_3CH_2CH_3$
Butane	C_4H_{10}	$CH_3(CH_2)_2CH_3$
Pentane	C_5H_{12}	$CH_3(CH_2)_3CH_3$
Hexane	C_6H_{14}	$CH_3(CH_2)_4CH_3$
Heptane	C_7H_{16}	$CH_3(CH_2)_5CH_3$
Octane	C_8H_{18}	$CH_3(CH_2)_6CH_3$

4 Complete the following table concerning some simple alkenes.

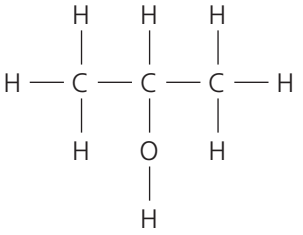
Name	Molecular formula	Condensed structural formula
Ethene	C_2H_4	$CH_2=CH_2$
Propene	C_3H_6	$CH_3CH=CH_2$
But-1-ene	C_4H_8	$CH_3CH_2CH=CH_2$
But-2-ene	C_4H_8	$CH_3CH=CHCH_3$
Pent-1-ene	C_5H_{10}	$CH_3CH_2CH_2CH=CH_2$
Pent-2-ene	C_5H_{10}	$CH_3CH_2CH=CHCH_3$

5 Complete the following table concerning some simple alkanols.

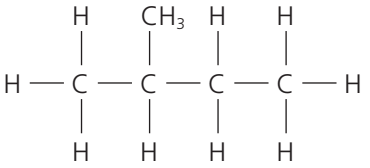
Name	Molecular formula	Condensed structural formula
Methanol	CH_3OH	CH_3OH
Ethanol	C_2H_5OH	CH_3CH_2OH
Propan-1-ol	C_3H_7OH	$CH_3CH_2CH_2OH$
Propan-2-ol	C_3H_7OH	$CH_3CH(OH)CH_3$
Butan-1-ol	C_4H_9OH	$CH_3CH_2CH_2CH_2OH$
Butan-2-ol	C_4H_9OH	$CH_3CH_2CH(OH)CH_3$

6 The question is about the compounds A to F shown below.

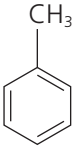
A



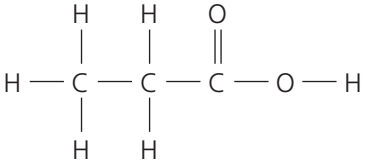
B



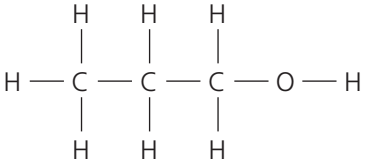
C



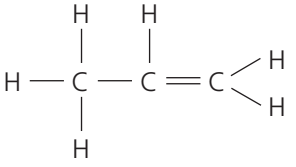
D



E



F



a) Which one is a saturated hydrocarbon?
B

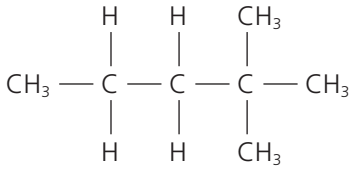
b) Which one is an aromatic compound?
C

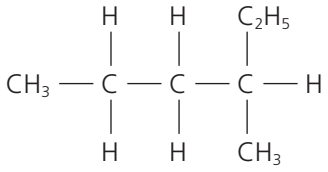
c) Which one is an alkene?
F

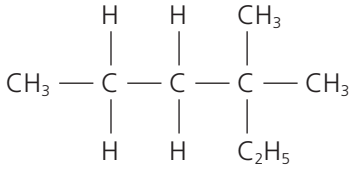
d) Which two are members of the same homologous series?
A and E

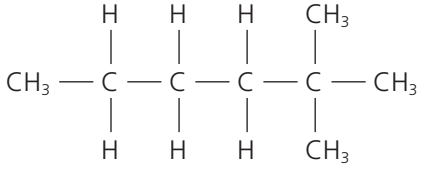
Part II Multiple choice questions

7 Which of the following is the structural formula of 3,3-dimethylhexane? Refer to page T24 for explanation.

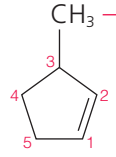
A 

B 

(C) 

D 

8 What is the IUPAC name of the following compound?



CH_3 — methyl group attached to the third carbon atom

∴ the IUPAC name of the compound is 3-methylcyclopentene.

A 3-methylcyclobutene
B 5-methylcyclobutene
(C) 3-methylcyclopentene
D 5-methylcyclopentene

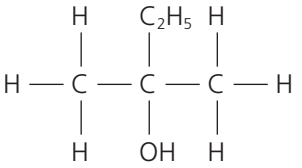
9 The correct name for the compound shown below is Refer to page T24 for explanation.



A 2-methylbut-3-ene.
B 3-methylbut-2-ene.
C 3-methylbut-3-ene.
(D) 2-methylbut-2-ene.

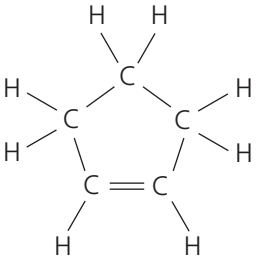
(Edexcel Advanced Subsidiary GCE, Unit 1, Jan. 2011, 10)

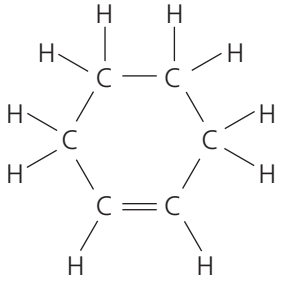
10 What is the IUPAC name of the following compound? Refer to page T24 for explanation.

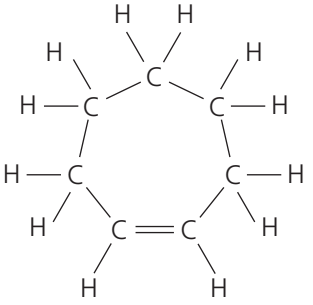


A 2-ethylpropan-1-ol
B 2-methylpropan-2-ol
C 2-methylbutan-1-ol
(D) 2-methylbutan-2-ol

11 The structures of three cycloalkenes are shown below:





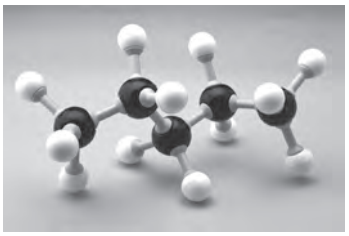


Cycloalkenes can be represented by a general formula. Which of the following is the general formula for cycloalkenes? (In these formulae, *n* is an integer greater than 2.)

A C_nH_{2n-4}
(B) C_nH_{2n-2}
C C_nH_{2n}
D C_nH_{2n+2}

(HKDSE, Practice paper 1A, 2012, 10)

12 The diagram below shows the molecular model of a compound.



key:
● carbon atom
○ hydrogen atom

Which of the following statements concerning the compound is INCORRECT?

A It is an aliphatic hydrocarbon.
B It is found in petrol.
C It is an alkane.
(D) Its structural formula is C_5H_{12}.

C_5H_{12} is the MOLECULAR FORMULA of the compound.

13 Which of the following statements about members of the alkane series is correct?

A The members have the same structural formula.
(B) The relative molecular mass of each successive member differs by 14.
C The volatility of the members increases with relative molecular mass.
D The boiling point of the members decreases with increasing number of carbon atoms per molecule.

14 Which of the following has the highest boiling temperature?

A Pentane, $CH_3CH_2CH_2CH_2CH_3$
(B) Hexane, $CH_3CH_2CH_2CH_2CH_2CH_3$
C 2-methylbutane, $CH_3CH(CH_3)CH_2CH_3$
D 2-methylpentane, $CH_3CH(CH_3)CH_2CH_2CH_3$

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2010, 6)

15 The difference in boiling temperature between methane (boiling point 109 K) and ethane (boiling point 185 K) is best explained by the different numbers of

A protons.
(B) electrons.
C covalent bonds.
D hydrogen bonds.

The strength of van der Waals' forces in alkanes increases with the number of electrons in the molecule.

(Edexcel Advanced Subsidiary GCE, Unit 2, Jan. 2012, 3)

16 Which of the following substances are sources of organic chemicals?

(1) Wood
(2) Rock
(3) Crude oil

A (1) and (2) only
(B) (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(HKCEE, Paper 2, 2008, 27)

17 Which of the following statements about 2-methylpentane are correct?

(1) It is soluble in water.
(2) It is a saturated hydrocarbon.
(3) It has the same molecular formula as hexane.

A (1) and (2) only
B (1) and (3) only
(C) (2) and (3) only
D (1), (2) and (3)

(1) 2-methylpentane is insoluble in water.
(3) Both 2-methylpentane and hexane have the molecular formula C_6H_{14}.

18 Which of the following statements concerning the compounds C_2H_4, C_3H_8 and C_4H_{10} is / are correct?
Refer to page T24 for explanation.

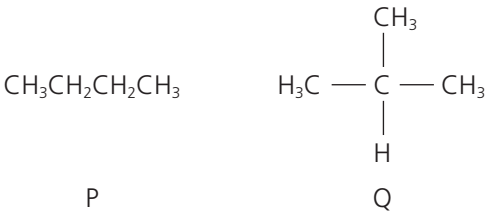
(1) They are gases at room temperature and pressure.
(2) They are unsaturated hydrocarbons.
(3) They have the same physical properties.

(A) (1) only
B (2) only
C (1) and (3) only
D (2) and (3) only

13 Option A — The members have the same general formula.
Option B — Each successive member in a homologous series differs from the next by a $-CH_2-$ unit, i.e. the relative molecular mass differs by 14.
Option C — The volatility of the members decreases with relative molecular mass.
Option D — The boiling point of the members increases with increasing number of carbon atoms per molecule.

14 Options B and D — Hexane molecule has a linear shape while 2-methylpentane has a more spherical and compact shape. The molecular shape of hexane allows greater surface contact between molecules. Thus, the van der Waals' forces in hexane are stronger than those in 2-methylpentane. Thus, the boiling point of hexane is higher.

19 The structures of two carbon compounds, P and Q, are shown below:



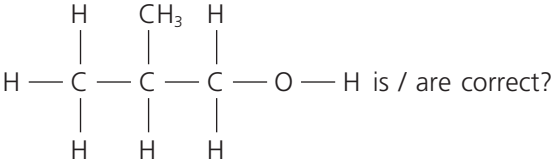
Which of the following statements concerning the compounds are correct?

(1) They have the same general formula.
(2) The boiling point of P is higher than that of Q.
(3) P dissolves in water readily but Q does not.

(A) (1) and (2) only
B (1) and (3) only
C (2) and (3) only
D (1), (2) and (3)

(1) Both P and Q are alkanes. Thus, they have the same general formula C_nH_{2n+2}.
(3) Both P and Q are insoluble in water.

20 Which of the following statements about

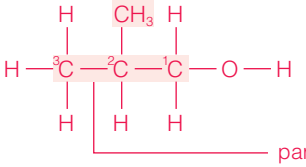


is / are correct?

(1) Its IUPAC name is methylpropanol.
(2) Its general formula is $C_nH_{2n+1}OH$.
(3) The $-CH_3$ group is the functional group.

A (1) only
(B) (2) only
C (1) and (3) only
D (2) and (3) only

(1)



methyl group attached to the second carbon atom, it is not necessary to specify the location of the methyl group as there is no other combination for a methyl group attached to a three-carbon parent chain.

parent chain containing 3 carbon atoms, with the –OH group attached to the first carbon atom, i.e. propan-1-ol

∴ the IUPAC name of the compound is methylpropan-1-ol.
(3) The $-CH_3$ group is NOT a functional group.

Part III Structured questions

Answers for the questions of the public examinations in Hong Kong are not provided.

21 Write the structural formulae of the following compounds:

Refer to page T25.

a) methylpropane

b) butane

c) 2,4-dimethylpentane

d) 2-methylbut-1-ene

e) 3-methylhex-2-ene

f) butan-2-ol

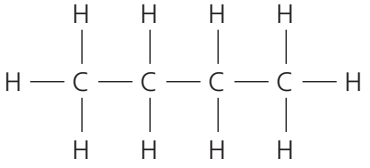
g) 2-methylpentan-2-ol

h) butanoic acid

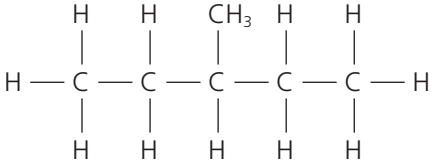
i) cyclohexanol

22 Write the condensed structural formulae of the following compounds.

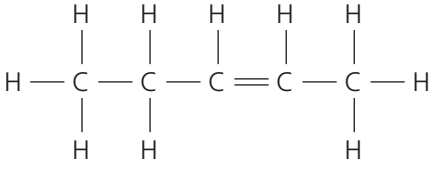
a)



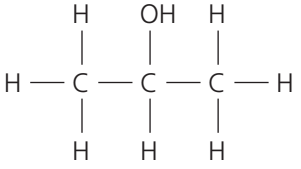
b)



c)



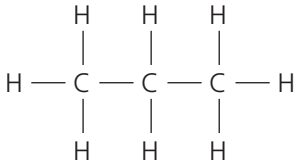
d)



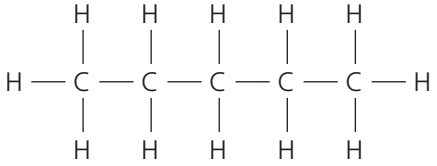
a) $CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$
b) $CH_3CH_2CH(CH_3)CH_2CH_3$
c) $CH_3CH_2CH{=}CHCH_3$
d) $CH_3CH(OH)CH_3$

23 Name the following compounds by the IUPAC system.

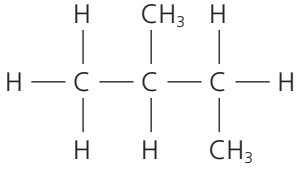
a)



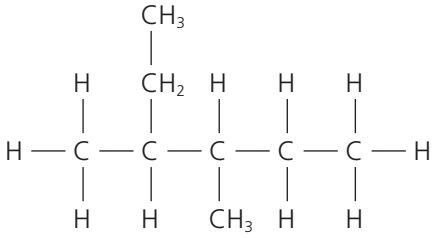
b)



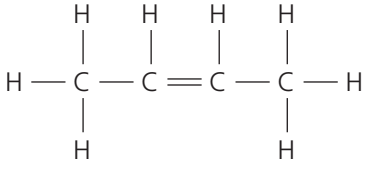
c)



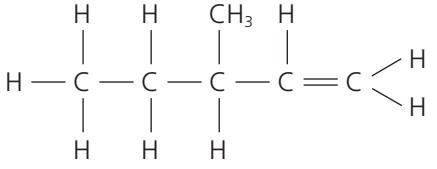
d)



e)



f)



a) propane
b) pentane
c) 2-methylbutane
d) 3,4-dimethylhexane
e) but-2-ene
f) 3-methylpent-1-ene

24 Methane is the first member of the homologous series of alkanes.
Refer to page T25.

a) Write the general formula for an alkane.

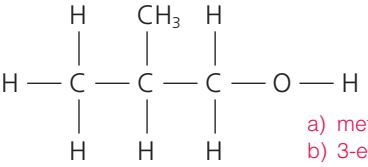
b) Draw a diagram to show the shape of a methane molecule, using lines to represent covalent bonds. Name the shape that a methane molecule adopts.

c) State how the chemical properties and a named physical property of members of a homologous series change, if at all, as the relative molecular mass increases.

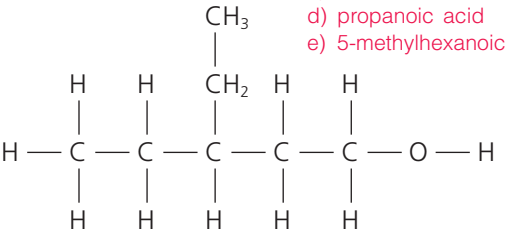
(Edexcel GCE O Level, Paper 2, May 2010, 2(a))

25 Name the following compounds by the IUPAC system.

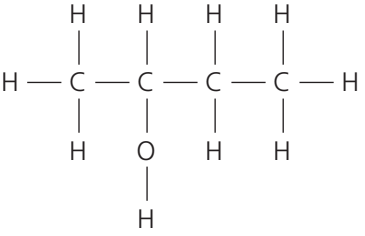
a)



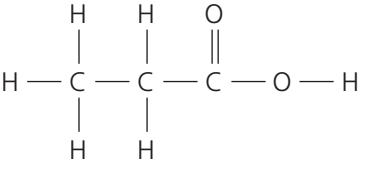
b)



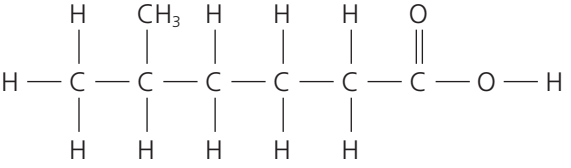
c)



d)



e)



a) methylpropan-1-ol
b) 3-ethylpentan-1-ol
c) butan-2-ol
d) propanoic acid
e) 5-methylhexanoic acid

26 The alkanes are a homologous series of hydrocarbons. The table shows information about some straight-chain alkanes. Refer to page T26.

Alkane	Molecular formula	Boiling point (°C)
Methane	CH_4	–162
Ethane	C_2H_6	–89
Propane	C_3H_8	–42
Butane	C_4H_{10}	–0.5

a) What is meant by a 'homologous series'?

b) Explain why the boiling points increase down the alkane homologous series.

(OCR Advanced Subsidiary GCE, Chem. A, F322, Jun. 2010, 1(a))

Unit Exercise

27 The formulae C_2H_6 and C_3H_8 represent two carbon compounds.

a) State why these compounds are described as

i) hydrocarbons;
They contain atoms of hydrogen and carbon only.

ii) saturated.
All the atoms are connected by single bonds.

b) The compounds C_2H_6 and C_3H_8 are members of the same homologous series.

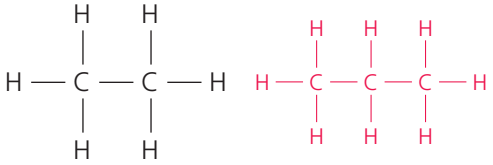
i) What is the name of this homologous series?
Alkane

ii) What is the general formula of this homologous series?
C_nH_{2n+2}

iii) Other than having the same general formula, state TWO other characteristics of members of the same homologous series.

iii) Any two of the following:
- Each member in a homologous series differs from the next by a $-CH_2-$ unit.
- They show a gradual change in physical properties.
- They show similar chemical properties.

c) The structural formula of C_2H_6 is

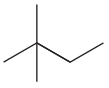


Write the structural formula of C_3H_8.

(Edexcel IGCSE (Higher Tier), Paper 2H, Nov. 2010, 3(a)–(c))

28 Petrol is a blend of many different compounds, most of which are hydrocarbons.

a) Complete the table below showing some of the hydrocarbons found in petrol.

Hydrocarbon	Molecular formula	Skeletal formula	Aliphatic hydrocarbon or aromatic hydrocarbon?	IUPAC name
W	C_6H_6		aromatic hydrocarbon	benzene
X	C_6H_{12}		aliphatic hydrocarbon	cyclohexane
Y	C_6H_{14}		aliphatic hydrocarbon	hexane
Z	C_6H_{14}		aliphatic hydrocarbon	2,2-dimethylbutane

b) Would you expect hydrocarbon Y to have a higher boiling point than hydrocarbon Z? Explain your answer.
Refer to page T26.

c) Decane ($C_{10}H_{22}$) is another hydrocarbon found in petrol. Would you expect decane or hydrocarbon Y to be more viscous? Explain your answer.
Refer to page T26.